

The Feynman Notes

A Comprehensive Study Guide to Physics Through Feynman's
Eyes

*Compiled from The Feynman Lectures on Physics, QED, The Character of Physical Law, Surely
You're Joking Mr. Feynman, and archival interviews.*

RAG Knowledge Source — Feynman Digital Twin Project

*Richard P. Feynman (1918–1988)
Nobel Laureate in Physics, 1965*

"Physics is like sex: sure, it may give some practical results, but that's not why we do it."

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Chapter 1: Who Was Richard Feynman?

Richard Phillips Feynman was born on May 11, 1918, in Queens, New York. He is widely regarded as one of the most brilliant and influential physicists of the 20th century. He received the Nobel Prize in Physics in 1965, jointly with Sin-Itiro Tomonaga and Julian Schwinger, for his work on quantum electrodynamics (QED). But awards alone cannot capture who Feynman was.

Feynman grew up in Far Rockaway, Queens, where his father, Melville, cultivated in him a deep curiosity about the natural world. From an early age, Feynman was taught not merely to name things but to understand them. His father would take him on walks and say: 'See that bird? In Portuguese it is called thus. In Chinese it's called thus. In Japanese it's called thus. Now you know the name of that bird in many languages — but you know absolutely nothing about the bird.' This lesson — that knowing the name of something is not the same as understanding it — became the philosophical bedrock of Feynman's entire scientific career.

Early Life and Education

Feynman attended MIT for his undergraduate studies, graduating in 1939, and then earned his PhD from Princeton University in 1942. His doctoral thesis, supervised by John Archibald Wheeler, introduced the path integral formulation of quantum mechanics — a reformulation so radical that it changed how physicists think about the quantum world.

During World War II, Feynman worked on the Manhattan Project at Los Alamos under J. Robert Oppenheimer. He was known for his irreverence, his lock-picking hobby (to the frustration of security officers), and his ability to cut through complex problems to their essential core. The experience left him with both great pride and profound moral ambiguity.

Teaching at Caltech

Feynman taught at the California Institute of Technology (Caltech) from 1950 until his death in 1988. His undergraduate physics lectures from 1961 to 1963 were compiled into *The Feynman Lectures on Physics*, a three-volume set that remains the most widely read physics textbook in history. The lectures are notable not just for their depth but for their insistence on physical intuition over mathematical formalism.

"When I was in high school, I had this notion that the way to learn a subject was not to just read the book, but to go off and think about it yourself — to work it out yourself, to try to discover everything that everyone had discovered before you." — Richard Feynman

Personal Character

Feynman was famous for his unconventional personality. He played bongo drums in nightclubs. He cracked safes at Los Alamos for amusement. He worked on physics problems in strip clubs because the noise and the drinks helped him concentrate. He was married three times. He learned to draw and sold portraits under the pseudonym 'Ofey'. He was a safe-cracker, a storyteller, a painter, and above all

a teacher who believed the highest calling was to make the complex simple.

Chapter 2: The Feynman Technique — Learning Anything

The Feynman Technique is a four-step mental model for learning any concept deeply. Feynman never wrote it down as a formal technique — it was distilled from his personal notebooks and his advice to students. The technique exploits a simple truth: if you cannot explain something simply, you do not understand it.

Step 1: Choose a concept: Pick any concept you want to understand. Write its name at the top of a blank page.

Step 2: Teach it to a child: Explain the concept in simple language, as if teaching a 12-year-old. No jargon. No shortcuts. Use analogies.

Step 3: Identify gaps: Where your explanation breaks down — where you reach for jargon, where you stall — is where your understanding is weakest. Go back to the source material.

Step 4: Simplify and use analogies: Return and re-explain. Replace technical terms with simple ones. Use stories and analogies. Repeat until the explanation is smooth, honest, and clear.

"If you cannot explain something in simple terms, you don't understand it." — Richard Feynman

Why It Works

The technique forces active recall rather than passive re-reading. It externalizes your understanding, making gaps visible. The act of simplifying forces you to confront circular reasoning and borrowed language. Feynman applied this to himself: whenever he could not reconstruct a result from scratch, he put it on his 'to-learn' list and worked it through until the derivation felt inevitable.

Application to Physics

Feynman applied this approach to every area of physics he taught. When introducing quantum mechanics to undergraduates, he did not begin with the Schrodinger equation — he began with the double-slit experiment, building up the strangeness of quantum behaviour from direct observation before introducing formalism. The equation came after the intuition, never before.

Chapter 3: Quantum Mechanics — The Strange World of the Very Small

Quantum mechanics is the branch of physics that describes the behaviour of matter and energy at the smallest scales — atoms, electrons, photons, and subatomic particles. Feynman had a famous warning about quantum mechanics:

"I think I can safely say that nobody understands quantum mechanics. Do not keep saying to yourself, if you can possibly avoid it, 'but how can it be like that?' because you will go down the drain into a blind alley from which nobody has yet escaped. Nobody knows how it can be like that." — Richard Feynman

The Double-Slit Experiment

Feynman called the double-slit experiment 'the only mystery of quantum mechanics.' Here is the essential setup: a source fires particles — electrons, photons, even buckyballs — at a barrier with two narrow slits. Behind the barrier is a detection screen.

Classical expectation: the particles should pile up in two bands behind the two slits, like bullets fired through two gaps in a wall.

Quantum reality: the particles produce an interference pattern — alternating bands of high and low probability — as if each particle passes through both slits simultaneously and interferes with itself.

The stranger twist: if you place a detector at the slits to find out which slit each particle goes through, the interference pattern disappears. The act of measurement collapses the quantum behaviour.

Feynman's interpretation: 'The electron does anything it likes. It just goes in any direction at any speed, forward or backward in time, however it likes — and then you add up the amplitudes.'

Wave Functions and Probability

In quantum mechanics, the state of a particle is described by a wave function, typically written as the Greek letter psi (the symbol ψ). The wave function is a complex-valued function of position and time. Its physical meaning was given by Max Born:

The probability of finding a particle at position x at time t is proportional to $|\psi(x,t)|^2$ — the square of the absolute value of the wave function.

The wave function itself is not directly observable. Only its squared magnitude corresponds to a measurable probability. This is the Born rule, and it is one of the postulates of quantum mechanics that cannot be derived from anything deeper — it is taken as a fundamental axiom of the theory.

The Schrodinger Equation

The time evolution of the wave function is governed by the Schrodinger equation. In its time-dependent form it states that the rate of change of ψ with respect to time equals (i times $\hbar$ divided by $2m$)

times the second spatial derivative of ψ , plus the potential energy $V(x,t)$ times ψ .

$$i \cdot \hbar \cdot \left(\frac{d}{dt} \right) \psi(x,t) = \left[-\left(\frac{\hbar^2}{2m} \right) \left(\frac{d^2}{dx^2} \right) + V(x,t) \right] \cdot \psi(x,t)$$

Here, $\hbar$ is the reduced Planck constant (h divided by 2π), m is the particle's mass, and $V(x,t)$ is the potential energy at position x and time t . The left-hand side is the time derivative; the right-hand side is the Hamiltonian operator acting on the wave function.

Superposition and Entanglement

Two of the most counterintuitive features of quantum mechanics:

Superposition: A quantum system can exist in multiple states simultaneously until it is measured. Schrodinger's famous cat is both alive and dead until observed.

Entanglement: Two particles can be correlated in such a way that measuring one instantly determines the state of the other, regardless of the distance between them. Einstein called this 'spooky action at a distance' and believed it proved quantum mechanics was incomplete. Bell's theorem and subsequent experiments proved Einstein wrong.

Heisenberg Uncertainty Principle

Feynman considered the uncertainty principle one of the most profound results in all of physics. It states that the position (x) and momentum (p) of a particle cannot both be known simultaneously with arbitrary precision:

$$\Delta(x) \cdot \Delta(p) \geq \hbar / 2$$

The uncertainty is not due to imperfect instruments. It is a fundamental property of nature. A particle does not have a definite position and momentum at the same time — these properties are not simultaneously real. Feynman explained this via the single-slit experiment: narrowing the slit to pin down a particle's vertical position causes its vertical momentum to spread out and become uncertain.

Chapter 4: Quantum Electrodynamics (QED)

Quantum Electrodynamics — QED — is the quantum field theory of electromagnetism. It describes how light and matter interact at the quantum level. It is the most precisely tested theory in the history of science: its predictions match experiment to more than ten decimal places. Feynman, Schwinger, and Tomonaga independently developed the mathematical framework for QED in the late 1940s, earning the 1965 Nobel Prize.

The Core Idea

All electromagnetic phenomena — light, electric fields, magnetic fields, chemical bonds, friction, the fact that you don't fall through the floor — arise from a single underlying process: electrons exchanging photons.

In QED, photons are the force carriers of electromagnetism. When two electrons repel each other, they do so by exchanging virtual photons — photons that exist only as intermediate states in the calculation and are never directly observed.

Path Integral Formulation

Feynman's great contribution was the path integral formulation of quantum mechanics, which provides the foundation for QED. The core idea: a quantum particle travelling from point A to point B does not take a single definite path. It takes all possible paths simultaneously. Each path contributes a complex amplitude. You sum (integrate) all these amplitudes. The squared magnitude of the total amplitude gives the probability.

This is not metaphor. It is the calculational procedure that produces correct predictions. In practice, paths close to the classical trajectory contribute amplitudes that add constructively. Wildly deviant paths mostly cancel. This is why large objects follow classical trajectories: the quantum interference selects the classical path.

Feynman Diagrams

Feynman invented a graphical notation for organising the complex perturbative calculations in QED. A Feynman diagram is a pictorial representation of a mathematical expression describing the behaviour of subatomic particles.

Straight lines represent fermions (electrons, positrons).

Wavy lines represent photons (the force carrier).

Vertices (where lines meet) represent interactions. Each vertex contributes a factor of the coupling constant e (elementary charge).

More complex diagrams (more vertices) contribute smaller corrections. The expansion in powers of the fine-structure constant (approximately $1/137$) converges rapidly.

The Fine-Structure Constant

The fine-structure constant, α , is the dimensionless coupling constant of QED. Its approximate value is $1/137$ (more precisely, approximately 0.0072973525693). It characterises the strength of the electromagnetic force. Feynman was obsessed with it:

"It has been a mystery ever since it was discovered more than fifty years ago, and all good theoretical physicists put this number up on their wall and worry about it. Immediately you would like to know where this number comes from... Nobody knows. It's one of the greatest damn mysteries of physics: a magic number that comes to us with no understanding by man." — Richard Feynman

Renormalisation

Early attempts at QED produced infinite results for physically measurable quantities — an apparent catastrophe. The resolution is renormalisation: a mathematical procedure for absorbing the infinities into redefined physical constants (mass and charge). While Feynman helped develop renormalisation and used it masterfully, he was always uncomfortable with it:

"The shell game that we play to find n , m , and j is technically called renormalisation. But no matter how clever the word, it is what I would call a dippy process! Having to resort to such hocus-pocus has prevented us from proving that the theory of quantum electrodynamics is mathematically self-consistent." — Feynman

Chapter 5: Classical Mechanics and the Laws of Motion

Classical mechanics, the physics of everyday motion, was Feynman's starting point for all physics education. He insisted students understand Newton's laws not as formulas to apply, but as descriptions of how nature actually behaves.

Newton's Three Laws

First Law (Inertia): An object at rest stays at rest; an object in motion stays in motion at constant velocity — unless acted upon by a net external force. Feynman's gloss: there is no physical difference between being at rest and moving at constant velocity. Both are the same state: no net force.

Second Law: Force equals mass times acceleration: $F = ma$. This is the equation of motion. Given the forces on an object and its initial conditions, this equation determines all future motion.

Third Law: For every action there is an equal and opposite reaction. Forces always come in pairs. The Earth pulls you down; you pull the Earth up. The magnitudes are equal; the accelerations differ because the masses differ.

Energy and Conservation Laws

Feynman considered conservation laws the deepest results in classical physics. The three great conservation laws are energy, momentum, and angular momentum. Each is a consequence of a symmetry of nature (Noether's theorem):

Energy conservation follows from time-translation symmetry — the laws of physics do not change over time.

Momentum conservation follows from spatial-translation symmetry — the laws of physics are the same everywhere.

Angular momentum conservation follows from rotational symmetry — the laws of physics are the same in every direction.

Least Action Principle

Feynman was particularly moved by the principle of least action, which he considered the most beautiful formulation of classical mechanics. Every physical path taken by a system is the one that minimises (or more precisely, makes stationary) the action S , defined as the integral of the Lagrangian $L = (\text{kinetic energy}) - (\text{potential energy})$ over time.

$$S = \text{integral of } L \, dt = \text{integral of } (T - V) \, dt$$

This variational principle contains all of classical mechanics and generalises elegantly to quantum mechanics via Feynman's path integral (where all paths are taken, weighted by e raised to the power $iS/\hbar$).

Chapter 6: Electromagnetism

In The Feynman Lectures, Volume II is devoted entirely to electromagnetism — a testament to how important Feynman considered it. He called Maxwell's equations one of the greatest achievements in all of physics.

Maxwell's Equations (Conceptual)

Maxwell's four equations — in their full differential form — unify electricity and magnetism into electromagnetism and predict the existence of electromagnetic waves travelling at the speed of light. Feynman always taught the physics before the mathematics.

Gauss's Law (Electric): The electric flux through any closed surface equals the enclosed charge divided by the permittivity of free space. Electric field lines begin on positive charges and end on negative charges.

Gauss's Law (Magnetic): The magnetic flux through any closed surface is zero. There are no magnetic monopoles. Magnetic field lines always form closed loops.

Faraday's Law: A changing magnetic field induces a circulating electric field. This is the basis of electric generators and transformers.

Ampere-Maxwell Law: A changing electric field (and electric current) produces a circulating magnetic field. Maxwell added the displacement current term, predicting electromagnetic waves.

"From a long view of the history of mankind, seen from, say, ten thousand years from now, there can be little doubt that the most significant event of the 19th century will be judged as Maxwell's discovery of the laws of electrodynamics." — Feynman

Chapter 7: Thermodynamics and Statistical Mechanics

Feynman's approach to thermodynamics was characteristically irreverent. Rather than beginning with abstract laws, he started with atoms — the physical picture that makes the abstract laws inevitable.

"If, in some cataclysm, all of scientific knowledge were to be destroyed, and only one sentence passed on to the next generations of creatures, what statement would contain the most information in the fewest words? I believe it is the atomic hypothesis (or the atomic fact, or whatever you wish to call it) that all things are made of atoms — little particles that move around in perpetual motion, attracting each other when they are a little distance apart, but repelling upon being squeezed into one another. In that one sentence, you will see, there is an enormous amount of information about the world, if just a little imagination and thinking are applied." — Feynman

The Three Laws of Thermodynamics

Zeroth Law: If A is in thermal equilibrium with B, and B is in thermal equilibrium with C, then A is in thermal equilibrium with C. This defines temperature.

First Law: Energy is conserved. The change in internal energy of a system equals the heat added minus the work done by the system.

Second Law: The entropy of an isolated system never decreases. Heat flows spontaneously from hot to cold, never the reverse. Entropy always increases or stays the same.

Third Law: The entropy of a perfect crystal at absolute zero temperature is zero. You cannot reach absolute zero in a finite number of steps.

Entropy and Disorder

Feynman explained entropy with characteristic clarity: entropy is a measure of the number of microscopic configurations (microstates) consistent with the observed macroscopic state. The Boltzmann formula gives entropy S as the product of Boltzmann's constant k and the natural logarithm of the number of microstates W :

$$S = k * \ln(W)$$

A glass of ice water has lower entropy than a glass of warm water because the ice arranges molecules in a more ordered crystalline structure — fewer possible microstates. The reason ice melts rather than warm water spontaneously freezing is statistical: the melted state has vastly more possible microscopic arrangements.

Chapter 8: Special Relativity

Special relativity is Einstein's theory describing how space and time are perceived differently by observers moving relative to each other. Feynman taught it as a logical consequence of two postulates, making it feel inevitable rather than miraculous.

Einstein's Two Postulates

Postulate 1 (Relativity): The laws of physics are the same in all inertial reference frames. There is no experiment that can distinguish 'being at rest' from 'moving at constant velocity'.

Postulate 2 (Speed of Light): The speed of light in a vacuum is the same for all observers, regardless of the motion of the source or observer. $c = 299,792,458$ metres per second exactly.

Time Dilation

A consequence of the postulates: a moving clock runs slower than a stationary one. If a clock moves at velocity v relative to an observer, the observer measures a longer time interval (t) between ticks than the proper time (t_0) measured by the clock itself:

$$t = t_0 / \sqrt{1 - v^2/c^2} = \gamma * t_0$$

where γ (the Lorentz factor) $= 1 / \sqrt{1 - v^2/c^2}$, and is always greater than or equal to 1. At everyday speeds γ is essentially 1 (no measurable dilation). At 99% the speed of light, γ is approximately 7.09 — a clock runs more than 7 times slower.

Length Contraction

A moving object is measured to be shorter along its direction of motion. If an object has proper length L_0 at rest, a moving observer measures its length L as:

$$L = L_0 / \gamma = L_0 * \sqrt{1 - v^2/c^2}$$

Mass-Energy Equivalence

Perhaps the most famous equation in physics follows directly from special relativity. The total energy E of a particle with rest mass m_0 and momentum p is:

$$E^2 = (m_0 * c^2)^2 + (p*c)^2$$

For a particle at rest ($p = 0$), this reduces to the iconic form: $E = m_0 * c^2$. Rest mass is a form of energy. This is not an analogy or a metaphor — mass and energy are genuinely interconvertible. Nuclear fission releases energy by converting a tiny fraction of nuclear mass into kinetic energy.

Feynman's insight: relativity does not change the laws of physics — it changes our understanding of space and time themselves. Space and time are not absolute; they are aspects of a four-dimensional spacetime that different observers slice differently.

Chapter 9: The Nature of Science and First Principles Thinking

Feynman had a fierce philosophy of science. He despised pseudoscience, cargo cult science, and intellectual dishonesty. He believed the job of a scientist was not to confirm theories but to try to disprove them.

"Science is the belief in the ignorance of experts." — Richard Feynman

What is Science?

Feynman's definition: science is the method by which we discover reliable knowledge about the natural world by testing ideas against experiment. An idea that cannot be tested is not science. An idea that refuses to be tested — because its proponents claim it is already known — is bad science.

Cargo Cult Science

In his 1974 Caltech commencement address, Feynman introduced the concept of cargo cult science: research that has the appearance of science — experiments, papers, statistics — but lacks the fundamental integrity of scientific thinking. He gave examples from psychology where results were never replicated, where controls were absent, where scientists fooled themselves.

"The first principle is that you must not fool yourself — and you are the easiest person to fool." — Richard Feynman

First Principles Thinking

Feynman always insisted on building up from first principles — the most basic, well-established facts — rather than reasoning by analogy. His method:

- Identify what you know for certain (experimentally verified facts).

- Identify what is assumption, model, or analogy.

- Build up the argument step by step, knowing exactly where each step comes from.

- Arrive at conclusions that are as solid as their premises — no more.

Applied to physics education: Feynman objected to students who could manipulate equations without understanding what the equations described. In his view, a formula without physical intuition was not knowledge — it was a tool you didn't know how to use.

Chapter 10: Famous Feynman Quotes and Teaching Philosophy

On Science and Knowledge

On understanding:

"I learned very early the difference between knowing the name of something and knowing something."

On curiosity:

"Study hard what interests you the most in the most undisciplined, irreverent and original manner possible."

On beauty:

"You can know the name of a bird in all the languages of the world, but when you're finished, you'll know absolutely nothing whatever about the bird... So let's look at the bird and see what it's doing — that's what counts."

On imagination:

"Physics is imagination in a straitjacket."

On teaching:

"If you can't explain something to a first year student, then you haven't really understood it."

On uncertainty:

"I think it's much more interesting to live not knowing than to have answers which might be wrong."

On nature:

"Nature uses only the longest threads to weave her patterns, so that each small piece of her fabric reveals the organization of the entire tapestry."

On doubt:

"I would rather have questions that can't be answered than answers that can't be questioned."

On fun:

"Hell, if I could explain it to the average person, it wouldn't have been worth the Nobel prize."

On knowing vs doing:

"What I cannot create, I do not understand."

Chapter 11: Key Equations and Derivations

This chapter provides the essential equations from each domain, in the form Feynman would present them — with physical meaning, not just symbols.

Newton's Second Law

$$F = m * a$$

Force equals mass times acceleration. The equation of classical motion.

Kinetic Energy

$$KE = (1/2) * m * v^2$$

Energy of motion. Depends on mass and the square of velocity.

Gravitational Potential Energy

$$PE = m * g * h$$

Energy stored by height in a gravitational field.

Coulomb's Law

$$F = k * q_1 * q_2 / r^2$$

Electrostatic force between two charges, inversely proportional to distance squared.

Schrodinger (time-dependent)

$$i\hbar(d/dt)\psi = H\psi$$

Governs how quantum wave functions evolve in time.

de Broglie Relation

$$\lambda = h / p$$

Every particle has a wavelength inversely proportional to its momentum.

Heisenberg Uncertainty

$$\Delta x * \Delta p \geq \hbar/2$$

Fundamental limit on simultaneous knowledge of position and momentum.

Mass-Energy Equivalence

$$E = m * c^2$$

Rest mass energy. Foundational result of special relativity.

Boltzmann Entropy

$$S = k * \ln(W)$$

Entropy as logarithm of the number of microscopic states.

Lorentz Factor

$$\gamma = 1 / \sqrt{1 - v^2/c^2}$$

The factor by which time dilates and length contracts at velocity v .

Fine Structure Constant

$$\alpha \approx e^2 / (4\pi\epsilon_0\hbar c) \approx 1/137$$

Dimensionless coupling constant of electromagnetism.

Photon Energy

$$E = h * f = \hbar * \omega$$

Energy of a photon is proportional to its frequency.

Chapter 12: Feynman Diagrams Explained

A Feynman diagram is a graphical tool invented by Feynman to organise the perturbative expansion of quantum field theory calculations. Each diagram represents a mathematical term in the expansion of the probability amplitude for a given physical process.

How to Read a Feynman Diagram

Time runs from left to right (or bottom to top, depending on convention).

Straight lines with arrows represent fermions. Arrow direction: forward in time = particle; backward in time = antiparticle.

Wavy lines represent photons (the electromagnetic force carrier).

Spiral (spring-like) lines represent gluons (the strong force carrier in QCD).

Dashed lines represent scalar particles (e.g., Higgs boson).

Vertices where lines meet represent interaction points. In QED each vertex contributes a factor of the coupling constant e .

The Simplest QED Process: Electron-Electron Scattering

Two electrons approach each other and exchange a single virtual photon. In the diagram: two incoming electron lines, a wavy photon line connecting them at two vertices, two outgoing electron lines. This is the lowest-order (tree-level) diagram. It contributes the dominant term in the amplitude.

Higher-Order Corrections

The perturbative expansion includes diagrams with more vertices. A one-loop diagram has an additional virtual particle loop and contributes a correction of order α (approximately $1/137$) relative to the tree level. A two-loop correction is of order α^2 . Because α is small, higher-loop corrections are progressively smaller — the expansion converges rapidly.

QED has been tested to the precision of the electron's anomalous magnetic moment ($g-2$). Theory and experiment agree to more than 10 decimal places — the most precise agreement between any theory and experiment in the history of science.

Chapter 13: The Challenger Disaster — Science Meets Reality

On January 28, 1986, the Space Shuttle Challenger broke apart 73 seconds after launch, killing all seven crew members. Feynman was appointed to the Presidential Commission investigating the disaster. His work on the commission exemplifies his scientific values: insistence on honesty, distrust of groupthink, and communication of risk.

The O-Ring Demonstration

During a televised commission hearing, Feynman produced a glass of ice water, a C-clamp, and a piece of O-ring material identical to that used in the shuttle's solid rocket booster joints. He compressed the material, submerged it in the ice water, and demonstrated that at low temperatures the rubber lost its elasticity and could not seal the joint properly.

The O-rings had been designed with a minimum operating temperature of 53 degrees Fahrenheit. The launch-day temperature at Kennedy Space Center was 28 degrees Fahrenheit — 25 degrees below the minimum design threshold. NASA had known there were problems with the O-rings at cold temperatures but had proceeded with the launch anyway.

Feynman's Appendix

Feynman insisted on appending his personal conclusions to the commission report, which NASA initially tried to suppress. In it, he found that NASA management's estimate of the probability of catastrophic failure was 1 in 100,000 — wildly inconsistent with the engineers' own estimates of 1 in 100 or 1 in 300.

"For a successful technology, reality must take precedence over public relations, for Nature cannot be fooled." — Richard Feynman, Appendix to the Rogers Commission Report

Chapter 14: Feynman's Approach to Problem Solving

Feynman's problem-solving approach was legendary among physicists. His colleague Murray Gell-Mann described watching Feynman attack a problem as 'like watching a fish swim in water — impossibly fluid, impossibly natural.'

The Feynman Algorithm (as described by colleagues)

Step 1: Write down the problem.

Step 2: Think very hard.

Step 3: Write down the solution.

This is both a joke and a profound truth. Feynman did not follow a mechanical algorithm. He visualised the physics, played with limiting cases, used dimensional analysis, and relied on physical intuition built over decades.

Practical Strategies Feynman Used

Dimensional analysis: Check that equations have consistent units on both sides. Wrong dimensions = definitely wrong answer.

Limiting cases: Set a parameter to zero or infinity. Does the equation reduce to something you know? This catches algebra errors and builds intuition.

Physical pictures: Before calculating, draw a picture. What is physically happening? The mathematics should describe a picture, not replace it.

Estimation: Feynman was a master of order-of-magnitude estimates. A quick back-of-the-envelope calculation prevents wasting time on approaches that are off by factors of millions.

Play: Feynman frequently solved problems while playing — drawing, drumming, walking. Unconscious processing is real. Take breaks.

If you are ever stuck on a problem, try explaining it to someone else. Usually by the time you have described the problem carefully enough for another person to understand it, you have solved it yourself. -- paraphrased from Feynman's teaching

Chapter 15: Practice Questions in Feynman's Style

Feynman's exams were legendary for testing understanding, not memorisation. These questions are in that spirit: conceptual, physical, and designed to reveal whether you understand or merely remember.

Conceptual Questions

1. A spaceship travels at 99% of the speed of light toward a planet 10 light-years away. How long does the journey take from the perspective of someone on the spaceship? From the perspective of someone on the planet?
2. Why does the interference pattern in the double-slit experiment disappear when you measure which slit the particle went through? Explain without using the word 'observation'.
3. If entropy always increases, how do living organisms create order? Are they violating the second law of thermodynamics?
4. The fine-structure constant is approximately $1/137$. What would happen to the universe if it were $1/100$ instead? What about $1/1000$?
5. Explain why you cannot reach the speed of light. Use the concept of relativistic mass — and then explain why physicists today prefer not to use the concept of relativistic mass.

Estimation Problems

6. Estimate the number of atoms in the human body. Show your reasoning.
7. The sun radiates approximately 3.8×10^{26} watts. Using $E = mc^2$, estimate how much mass the sun loses every second.
8. If you cooled a cup of water to absolute zero, how much of its initial thermal energy would you have removed? Estimate the energy in joules.

Derivation Questions

9. Derive the uncertainty principle for energy and time from the uncertainty principle for position and momentum.
10. Starting only from Newton's laws and the definition of kinetic energy, derive the work-energy theorem.

"The test of all knowledge is experiment. Experiment is the sole judge of scientific truth."

— Richard Feynman, *The Feynman Lectures on Physics*, Vol. I